Phase 1 Project Proposal: Problem Definition and Paper Selection

Wearable Biosignal-Based Cognitive Stress and Affective State Classifier

Project Metadata

Course Track: Track A (Problem Definition, Paper Selection, Data Setup)
Group ID: F26-02
Submission Milestone: Phase 1 Submission
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1 Executive Summary & Project Overview

This project addresses real-time affective state recognition by analyzing multimodal physiological biosignals collected from wearable sensors. By leveraging synchronized Electrocardiogram (ECG) and Electrodermal Activity (EDA) signals from the benchmark WESAD dataset, the project establishes a robust, lightweight machine learning and deep learning classification pipeline designed to accurately distinguish between relaxed baseline conditions and acute cognitive stress states.

2 Domain and Problem Statement

2.1 Domain Context

Affective computing and biomedical signal processing play a critical role in preventive health, adaptive interfaces, and automated mental health tracking. Continuous physiological monitoring provides objective biomarkers of stress, bypassing the subjectivity and lag associated with self-report questionnaires.

2.2 Problem Statement & Research Gap

While modern consumer wearables can continuously collect physiological streams, raw biosignals are high-dimensional, noisy, and computationally expensive to store and process. Many existing approaches rely on heavy clinical instrumentation or complex signal transformations that cannot scale to lightweight computational environments. A critical research gap exists in creating an end-to-end, reproducible pipeline that balances multimodal feature fusion (combining cardiac and electrodermal dynamics) with computational efficiency, while systematically comparing standard machine learning models against sequence-based deep learning architectures for subject-generalizable affect classification.

3 Dataset Specifications

Parameter	Specification Details
Dataset Name	WESAD (Wearable Stress and Affect Detection)
Link	https://www.kaggle.com/datasets/mohamedasem318/wesad-full-dataset/data
Subjects Included	15 active subjects (S2–S17; S1 and S12 excluded due to sensor malfunction)
Selected Modalities	RespiBAN Chest Sensors: Electrocardiogram (ECG @ 700 Hz) and Electrodermal Activity (EDA @ 700 Hz)
Target Classes	Binary Classification: Baseline / Relaxed (Label 1) vs. Cognitive Stress (Label 2)
Data Storage & Structure	Synchronized dictionary files (`.pkl`) containing time-aligned biosignal channels and synchronized protocol labels

4 Primary Reference Paper

- **Paper Title:** Introducing WESAD, a multimodal dataset for Wearable Stress and Affect Detection
- **Authors:** Philip Schmidt, Attila Reiss, Robert Duerichen, Claus Marberger, and Kristof Van Laerhoven
- **Publication Venue:** ACM International Conference on Multimodal Interaction (ICMI '18), 2018
- **DOI Link:** https://doi.org/10.1145/3242969.3242985

5 Clearly Defined Scope of the Project

5.1 In-Scope Activities

- **Data Preprocessing & Cleaning:** Parsing synchronized dictionary structures (`.pkl`), extracting chest ECG and EDA signals, filtering non-target protocol states, and exporting a lightweight CSV format for rapid collaborative iteration.
- **Feature Engineering & Baseline Modeling:** Extracting time-domain and frequency-domain cardiac and electrodermal features to train a baseline Random Forest classifier in Scikit-Learn.
- **Deep Learning Classification:** Developing 1D Convolutional Neural Networks (1D-CNN) and Long Short-Term Memory (LSTM) networks in PyTorch/TensorFlow to capture sequential biosignal dynamics.
- **Evaluation & Benchmarking:** Performing cross-subject model evaluation using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics, accompanied by error and bias checks.

5.2 Out-of-Scope Activities

- Deployment on live physical microcontroller/edge hardware in real time.
- Multi-class affect classification beyond relaxed and cognitive stress (e.g., secondary transient states, amusement, or meditation questionnaire sub-scores).